

Anomaly Detection in Systems using LogBERT with LLM-Powered Analysis

CSE-330 Machine Learning Sessional Project

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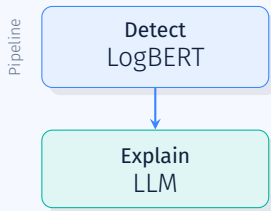
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Introduction

Modern systems generate **massive volumes of logs** that traditional rule-based monitoring cannot scale to handle.

Deep learning models such as **LogBERT** treat log sequences like natural language to identify deviations — but *detecting* an anomaly is only half the battle.

This project couples LogBERT's high-speed detection with **open-source LLMs** to deliver **human-readable root-cause analysis** and remediation guidance.



Motivation

The Problem

In high-stakes environments, downtime carries heavy economic and infrastructural cost. DevOps engineers must *manually correlate* anomalous logs — a slow, error-prone process.

Our Vision

Bridge the gap between **detection** and **resolution**:

- **LogBERT** flags anomalies
- **LLM** traces root causes and proposes fixes — automatically

Goal

Reduce mean-time-to-detect (MTTD), mean-time-to-repair (MTTR) through automated, explainable anomaly analysis

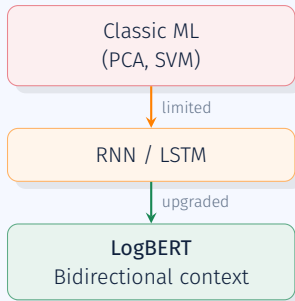
Problem Definition — Detection

Traditional ML

- Binary classification, PCA, one-class SVM
- Cannot capture *temporal* patterns in log sequences

RNN / LSTM

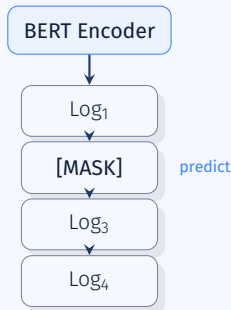
- Models sequential data
- Only left-context — misses bidirectional associations



Problem Definition — LogBERT

LogBERT [1] adapts BERT's *masked language modelling* to log sequences.

- Each log line template (*log key*) is embedded contextually
- Self-attention captures associations across the **full window**
- Trained to predict masked log keys — deviations signal anomalies



Problem Definition — Interpretation

The Remaining Gap

LogBERT tells us *that* something is wrong — but not *why*, nor *what to do*. In time-critical incidents this leaves engineers to correlate logs manually, slowing resolution.

Our Contribution

Pipe anomalous log sequences directly into an open-source **LLM** to obtain:

- Human-readable root-cause analysis
- Suggested mitigation strategies

Related Work

LogBERT [1]

Applies masked language modelling to log key sequences for unsupervised anomaly detection. Learns patterns of *normal* behaviour during training.

LLM-LADE [2]

Integrates large language models for *both* detection and explanation. Demonstrates that LLMs can reason about log anomalies in natural language.

Key Insight

Both approaches have complementary strengths that our pipeline unites

Limitations of Existing Works

LogBERT alone

- Detects anomalies — cannot explain them
- Manual log correlation still required, slowing incident response

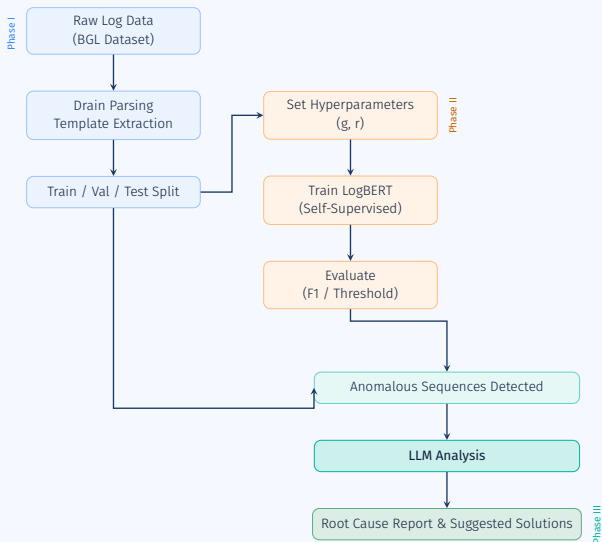
LLM-LADE

- Routes *all* logs through the LLM — very high inference cost
- Practical throughput too low for 7B–8B parameter models at scale

Our Fix

Pre-filter with LogBERT, send *only* anomalies to the LLM

Project Methodology



Methodology — Key Hyperparameters

g — Candidate Set Size

For each masked token, the model ranks all log keys by predicted probability. The **top- g** form the candidate set — inside means normal, outside means anomalous.

Ranked log keys by probability

Top- g Candidate Set

Key **inside** top- $g \Rightarrow$ Normal

Key **outside** top- $g \Rightarrow$ Anomalous

r — Anomalous Token Threshold

r is the **maximum number of anomalous tokens** allowed per sequence. Exceeding r flags the *entire sequence* as anomalous.

Count tokens outside top- g

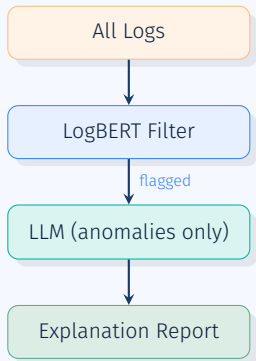
Count $\leq r \Rightarrow$ **Normal sequence**

Count $> r \Rightarrow$ **Anomalous sequence**

Our Novelty

In our pipeline, **LogBERT acts as a gatekeeper**: only flagged sequences reach the LLM.

- Drastically reduces LLM inference load
- Enables faster, targeted explanations
- Practical on commodity hardware with 7B–8B models



Results

Detection Performance (BGL)

Model	Prec.	Recall	F1
LogBERT (orig., 4.7M)	0.894	0.923	0.908
Ours (1M logs)	0.879	0.866	0.873

LLM Stage

- Generated root-cause explanations and remediation steps
- LLM verdict **aligned** with LogBERT on severe events (*rts panic!*, retransmission errors)

LLM Inference — Snapshots (1/2)

Running inference on anomalous log sequences

Inference Run Time

Inference Run 2

[illegible][illegible][illegible]

Note

Each run processes a **window of log keys** flagged as anomalous by LogBERT

Conclusion

- Hybrid framework: **LogBERT** for high-accuracy detection + **LLM** for explanation
 - LogBERT captures sequential log patterns with strong F1 on BGL
 - LLM provides **human-readable root-cause analysis** and mitigation guidance
 - Filtering reduces LLM load, improving **efficiency and scalability**
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- [1] H. Guo, S. Yuan, and X. Wu.

Logbert: Log anomaly detection via bert.

In 2021 International Joint Conference on Neural Networks (IJCNN), pages 1–8. IEEE, 2021.

- [2] Z. Zhang, S. Li, L. Zhang, J. Ye, C. Hu, and L. Yan.

Llm-lade: Large language model-based log anomaly detection with explanation.

Knowledge-Based Systems, 326:114064, 2025.